Question	Answer	Marks
1(a)(i)	direction of force on a (small test) mass **or** path in which a (small test) mass will move	**B1**
1(a)(ii)	(at surface,) lines (of force) are radial	**B1**
	Earth has large radius/height above surface is small so lines are (approximately) parallel	**B1**
	parallel lines → constant field strength	**B1**
1(b)	(change in) KE of rock = (change in) PE **or** $\frac{1}{2}mv^2 = GMm/R$	**C1**
	$(m)v^2 = (m)(2 \times 6.67 \times 10^{-11} \times 7.4 \times 10^{22}) / (1.7 \times 10^3 \times 10^3)$	**C1**
	$v = 2.4 \times 10^3\,\text{m s}^{-1}$	**A1**
	correct conclusion based on comparison of v with $2.8\,\text{km s}^{-1}$	**B1**
	or	
	(change in) KE of rock = (change in) PE	**(C1)**
	(at infinity) $E_P = (6.67 \times 10^{-11} \times 7.4 \times 10^{22} \times m) / (1.7 \times 10^3 \times 10^3)$ $= 2.9 \times 10^6\,m$	**(C1)**
	E_K of rock $= \frac{1}{2} \times m \times (2.8 \times 10^3)^2 = 3.9 \times 10^6\,m$	**(A1)**
	correct conclusion based on comparison of E_K and E_P values	**(B1)**
	or	

Question	Answer	Marks
	(change in) KE of rock = (change in) PE **or** $\frac{1}{2}mv^2 = GMm/R$	**(C1)**
	$(m)\,(2800)^2 = (m)\,(2 \times 6.67 \times 10^{-11} \times 7.4 \times 10^{22})\,/\,R$	**(C1)**
	$R = 1.3 \times 10^3$ km	**(A1)**
	correct conclusion based on comparison of R with 1.7×10^3 km	**(B1)**
	or	
	(change in) KE of rock = (change in) PE **or** $\frac{1}{2}mv^2 = GMm/R$	**(C1)**
	$(m)\,(2800)^2 = (m)\,(2 \times 6.67 \times 10^{-11} \times M)\,/\,(1.7 \times 10^6)$	**(C1)**
	$M = 1.0 \times 10^{23}$ kg	**(A1)**
	correct conclusion based on comparison of M with 7.4×10^{22} kg	**(B1)**

Question	Answer	Marks
3(a)	no intermolecular forces (so no potential energy)	B1